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Volatility and value-at-risk forecasting using BERT and transformer models incorporating investors' textual sentiments

Yuping Song [a,*], Yilun Zhang [b,1], Jiefei Huang [c], Aijun Yang [d,*]

[a] School of Finance and Business, Shanghai Normal University, Shanghai, China
[b] Research Center for Mathematics and Interdisciplinary Sciences, Shandong University, Qingdao, China
[c] School of Statistics and Information, Shanghai University of International Business and Economics, Shanghai, China
[d] College of Economics and Management, Nanjing Forestry University, Nanjing, China

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ABSTRACT

Effective risk management contributes to the stability of financial markets. This study investigates how decomposed investor sentiments affect financial volatility, emphasizing the role of sentiment polarity. Using BERT to extract positive and negative sentiment indices from stock forums, we find that negative sentiment has a stronger effect on realized volatility than positive sentiment, reflecting asymmetric investor reactions. Integrating these decomposed sentiment indices with technical indicators into a Transformer model significantly improves the forecasting of both realized volatility and value-at-risk (VaR). Our findings underscore the economic relevance of distinguishing sentiment polarity and offer novel insights into combining fine-grained sentiment features with deep learning for financial risk forecasting.

1. Introduction

With the expanding scale and increasing complexity of global financial markets, precise assessment and effective control of financial risks not only enlighten our understanding of financial market mechanisms but also hold significant practical value for financial market practices. Among various risk management tools, VaR for financial asset returns is extensively employed in academia and industry (Llacay and Peffer, 2017). Accurately estimating and predicting the volatility of financial assets is pivotal for VaR forecasting, which underscores its critical role in risk management (Bams et al., 2017).

Econometric models grounded in economic theory provide interpretable and stable parameter estimation for volatility forecasting. However, traditional approaches like GARCH and Stochastic Volatility models rely on low-frequency data and may overlook important intraday fluctuations. With the rise of high-frequency data, Andersen and Bollerslev (1998) proposed realized volatility as a more accurate measure of market fluctuations. Corsi (2009) later developed the heterogeneous autoregressive model of realized volatility (HAR-RV), which accounts for long memory and has become a standard modeling approach. Despite their advantages, econometric models mainly capture linear relationships and thus struggle to effectively handle nonlinear dynamics in increasingly complex financial markets (Dou et al., 2023). With improved computational power and GPU availability, deep learning models have become effective tools for modeling nonlinear financial data and time dependencies. Ito et al. (2022) used LSTM to forecast 1-minute FX returns

* Corresponding authors.
E-mail addresses: songyuping@shnu.edu.cn (Y. Song), ylzhang@mail.sdu.edu.cn (Y. Zhang), 24405002@suibe.edu.cn (J. Huang), ajyang81@163.com (A. Yang).
[1] Co-first Author

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from limit order book data, while Jing et al. (2024) applied an attention-enhanced Graph WaveNet to predict stock price crash risk in Chinese firms.

However, merely improving forecasting models may not significantly enhance predictive accuracy. Integrating relevant external structured and unstructured information, such as textual data converted into structured sentiment features, can effectively reinforce forecasting frameworks. For instance, Liu et al. (2024) used a BiLSTM with news-based textual features to forecast 1-day-ahead gold futures prices, while Liu et al. (2025) developed AT-FinGPT to predict financial risk by integrating earnings call audio and financial text. These studies underscore the critical role of sentiment-driven features in capturing latent market dynamics, notably investor sentiment, which profoundly influences market volatility through behavioral mechanisms such as Prospect Theory (Kahneman and Tversky, 1979) and herding behaviors (Bikhchandani et al., 1992).

Building upon the existing literature, this study innovatively incorporates decomposed positive and negative investor sentiment derived from stock forums using the BERT (Devlin et al., 2018) sentiment analysis model into volatility forecasting frameworks. Differentiating sentiment polarity provides a more comprehensive view of market participants' sentiment dynamics. In addition, this research introduces the Transformer (Vaswani et al., 2017) model, distinguished by its self-attention mechanism capable of capturing global dependencies in financial time series data. The empirical results indicate that the integration of decomposed sentiment indices significantly improves realized volatility forecasting accuracy, consequently enhancing VaR predictions. This study contributes by providing robust empirical evidence of the benefits derived from combining Transformer-based deep learning models with sentiment analysis methods, offering practical insights for risk management in financial markets.

The remainder of this paper is organized as follows: Section 2 outlines the core variables and modeling framework. Section 3 presents an in-depth discussion of the empirical results and analysis. Section 4 concludes the study.

2. Methodology

2.1. Realized volatility

This paper focuses on predicting realized volatility, a key metric in financial analysis for high-frequency data. Based on actual price movements, it accurately reflects market fluctuations. Following Andersen and Bollerslev (1998), it is measured as the sum of squared log returns using the following formula:

$$RV_d = \sum_{i=0}^{n} \left(\log \frac{P_{d,i+1}}{P_{d,i}} \right)^2, \tag{1}$$

where n represents the number of observations in a single day. For high-frequency data sampled every 5 min, $n = 48$. $P_{d,i}$ denotes the closing price at time i on day d.

2.2. Construction of sentiment indicators

For sentiment classification, FinBERT labels each comment as Positive or Negative. Inspired by Antweiler and Frank (2004), who used forum messages to construct bullish indicators for volatility prediction, we develop sentiment indicators based on view-weighted sentiment scores. These include the total sentiment indicator $SENT_t$, the positive sentiment indicator $SENT_{pos,t}$, and the negative sentiment indicator $SENT_{neg,t}$. By considering both the number and weight of comments, these indicators provide a balanced reflection of traders' sentiment expressions. Additionally, the inclusion of view-based weights captures differences in traders' attention, enabling a comprehensive depiction of market sentiment fluctuations. The formulas are as follows:

$$SENT_t = \left(\frac{M_t^{pos} - M_t^{neg}}{M_t^{pos} + M_t^{neg}} \right) \log \left(1 + M_t^{pos} + M_t^{neg} \right), \tag{2}$$

$$M_t^{pos} = \sum w_i s_i \mathrm{I}(l_i = Positive); M_t^{neg} = \sum w_i s_i \mathrm{I}(l_i = Negative), \tag{3}$$

$$SENT_{pos,t} = \left(\frac{M_t^{pos}}{M_t^{pos} + M_t^{neg}} \right) \log \left(1 + M_t^{pos} + M_t^{neg} \right), \tag{4}$$

$$SENT_{neg,t} = \left(\frac{M_t^{neg}}{M_t^{pos} + M_t^{neg}} \right) \log \left(1 + M_t^{pos} + M_t^{neg} \right), \tag{5}$$

where w_i represents the proportion of the i th comment's views to the total views on day t, s_i denotes the sentiment score predicted by FinBERT (ranging from [0,1]), and l_i is the sentiment label predicted by FinBERT. To ensure clear sentiment classification, for samples labeled as Neutral, the second-highest probability label is chosen as the final output, retaining only the Positive and Negative sentiment categories.

2.3. Research flow chart

This paper proposes a framework for predicting realized volatility, as shown in Fig. 1. The process is divided into three main stages. First, in the data processing and feature selection stage, the data is divided into interday features and intraday features. Descriptive statistical analysis (DSA) is used to assess the significance of the selected features.

Next, in the model prediction and evaluation stage, we test multiple deep learning and machine learning models, including Transformer, LSTM, SVR, etc. The models' predictive performance is compared, and the best model is selected based on effectiveness tests and the Diebold-Mariano (DM) test.

Finally, in the risk prediction stage, the selected model is used to calculate the VaR, which is validated through backtesting to evaluate its practical effectiveness in risk management.

3. Empirical analysis

3.1. Data source

This study employs both quantitative and textual data related to the Shanghai Composite Index (SSE Index). The trading data, comprising daily and 5-minute high-frequency records such as open, high, low, close prices, trading volume, and transaction amount, were obtained via the JoinQuant database (https://www.joinquant.com), a widely-used financial data provider for the Chinese market. The textual data were collected from the SSE Index forum on Eastmoney (http://guba.eastmoney.com/list,zssh000001.html), one of the largest online financial communities in China, known for its real-time investor discussions and sentiment expressions.

Using automated text data mining techniques, we extracted nearly a decade's worth of forum posts published between April 2013 and January 2022, each containing metadata such as publication timestamp, title, and full textual content. To enhance the dataset's quality and structure, we used Locality-Sensitive Hashing (LSH) to eliminate near-duplicate entries by detecting semantic similarities, and applied regular expressions to remove malformed characters and embedded HTML tags. These steps ensured a cleaner, more reliable textual dataset. Sentiment analysis using FinBERT is detailed in Appendix B.

After aligning the textual data with valid trading days (excluding weekends and public holidays), we retained 2151 trading days in total, covering 103,248 5-minute high-frequency intervals. Of these, 1890 days were used to train the forecasting models, and the

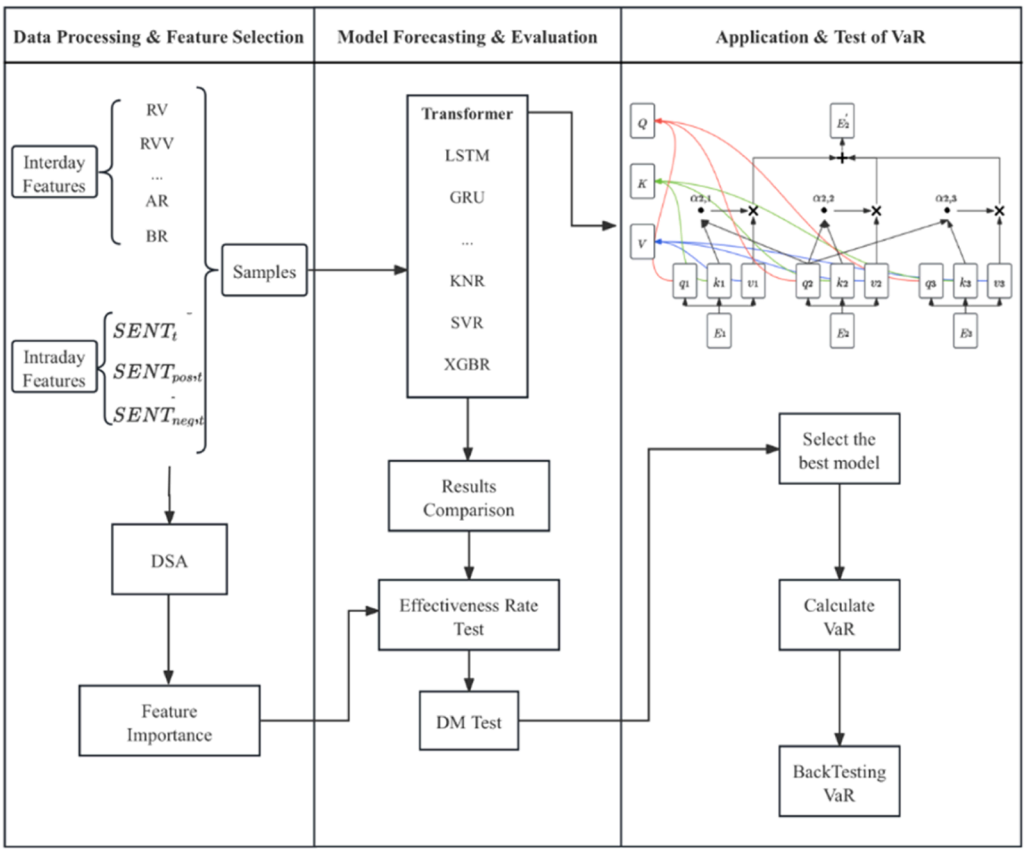

Fig. 1. Research flow chart.

remaining 261 days were reserved for out-of-sample testing. The final dataset comprises approximately 1.7 million sentiment-scored forum posts, which are used to construct daily sentiment indices as inputs for volatility and VaR prediction.

3.2. Descriptive statistical analysis

In terms of indicator selection, in addition to the intraday sentiment indicators constructed from textual information mentioned earlier, we also incorporated several interday feature indicators. The specific selection of these indicators was guided by the findings of Lei et al. (2021) on interday features. Detailed information on the selected features is provided in Table 1.

Table 2 demonstrates the importance of each variable in predicting the target volatility based on the Random Forest model. The calculated feature importance values highlight the relative contributions of the variables, providing a clear understanding of their roles in the model's predictions.

Fig. 2 depicts the trends of the overall sentiment indicator, the positive sentiment indicator, the negative sentiment indicator, and realized volatility. To better reveal the underlying trends of these indicators, we smooth the raw data using the Savitzky-Golay filter proposed by Press and Teukolsky, which fits a polynomial over a sliding window. This widely used method reduces noise and enhances trend identification in financial data.

This figure illustrates a clear positive correlation between the negative sentiment index and realized volatility, suggesting that pessimistic sentiment coincides with higher market volatility. During such periods, investors react more strongly to news, amplifying price fluctuations. In contrast, the positive sentiment index shows a weaker correlation with volatility, as optimism often reflects expectations of stability, leading to less trading and smaller price movements.

3.3. Realized volatility prediction

After classifying each post with FinBERT, we applied the sentiment quantification method from Section 2.2. By weighting each sentiment score by the post's share of daily views, we derived the overall, positive, and negative sentiment indices. FinBERT was chosen over general models for its ability to better capture financial language. Details on data normalization and tuning are in Appendix D.

We evaluated ten models using the metrics described in Appendix A. The first four are deep learning models, and the remaining six are traditional machine learning models, as shown in Table 3. Each model was tested with both the overall sentiment index and the decomposed positive and negative indices. To ensure robustness, we trained each model fifty times and used the average results for comparison. Final rankings are based on the average of five evaluation metrics. If multiple models had the same average rank, we further sorted them by total loss.

Table 3 shows that deep learning models clearly outperform traditional machine learning models in predicting realized volatility. This advantage stems from their ability to effectively capture the nonlinear structures and temporal dependencies that characterize financial markets. Among the evaluated deep learning models, which include WaveNet, GRU, LSTM and Transformer, the Transformer achieved the highest performance. Its superiority comes from the self-attention mechanism, which allows the model to access information from all positions in the input sequence and accurately capture long-range dependencies, outperforming causal convolution methods such as those used in WaveNet. In addition, across all ten models, those using the decomposed positive and negative sentiment indices consistently outperform those using only the overall sentiment index. Appendix Table D.1 lists the hyperparameter settings for the Transformer model.

We selected the top two models, Transformer and WaveNet, based on overall ranking from Table 3, both of which incorporate the decomposed positive and negative sentiment indices. Using the test set data, we plotted their true and predicted values as shown in Fig. 3. The Transformer model's prediction curve aligns more closely with the true values compared to WaveNet, which tends to oscillate mildly around the mean. This demonstrates the Transformer model's higher sensitivity to market fluctuations and its superior ability to forecast market trends.

Table 1
The Realized Volatility Forecasting Indicators.

	Name	Explanation
Interday Features	RVt-1	Lagged One-Day Realized Volatility
	RVVt-1	High-Frequency Trading Volume Volatility
	VOLt-1	Daily Transaction Volume
	CRt-1	(Closing price of the day - Closing price of the previous day)/Closing price of the previous day
	BIASt-1	(Closing price of the day - Average closing price of the previous 5-day)/Average closing price of the previous 5-day
	CDPt-1	(The highest price of the previous day + The lowest price of the previous day + 2 * Closing price of the previous day) / 4
	DMAt-1	Average closing price of the previous 5-day - Average closing price of the previous 10-day
	OIt-1	Opening price of the day - Closing price of the previous day
	ARt-1	(Closing price of the day - Opening price of the day) / (Opening price of the day - The lowest price of the day)
	BRt-1	(The highest price of the day - Closing price of the day) / (Closing price of the day - The lowest price of the day)
Intraday Features	SENTt	Overall Sentiment Index
	SENTpos,t	Positive Sentiment Index
	SENTneg,t	Negative Sentiment Index

Note: Descriptive statistical analysis for these features is shown in Appendix C.

Table 2
Variables importance via random forest model.

Variables	Importance
DMA	0.3751
OI	0.1635
bias	0.1354
volume	0.1232
RV_V	0.0724
neg_SENT	0.0577
pos_SENT	0.0202
BR	0.0179
CDP	0.0126
CR	0.0125
AR	0.0095

Fig. 2. Sentiment indices and RV.

Through the analysis of Table 3 and Fig. 3, we have validated the core argument of this study: the Transformer model incorporating decomposed positive and negative sentiment indices achieves optimal performance in predicting realized volatility. To further strengthen this validation, we designed ablation experiments and statistical tests.

To evaluate the impact of sentiment features from stock forum comments, we conducted ablation experiments by sequentially removing each sentiment feature from the data. We trained new models for each removal and evaluated their performance on the same test set. Using Eq. (6) and the metrics in Appendix A, we calculated error rates (ER) for each metric. Taking MSE as an example, $Loss_{drop}$ represents the MSE value of the model on the test set after removing a specific feature, while $Loss_{all}$represents the MSE value of the original model without any feature removal. Each experiment was repeated fifty times to ensure robustness, and the final results were averaged.

$$ER = \frac{Loss_{drop} - Loss_{all}}{Loss_{all}} \times 100\%, \tag{6}$$

Table 4 shows that for the optimal Transformer model, decomposed positive and negative sentiment features outperform the aggregated sentiment score across all five evaluation metrics. Among them, negative sentiment contributes more than positive sentiment, consistent with the trend in Fig. 2. This supports the view that negative comments more strongly influence investor behavior and market volatility, confirming the effectiveness and suitability of decomposed sentiment indicators.

We have already demonstrated empirically that incorporating decomposed sentiment indicators significantly improves model performance, particularly for the Transformer model. To statistically confirm this, we applied the DM test (Diebold and Mariano, 1995), which compares the performance of the Transformer model with the other nine models. The DM test results, presented in Table 5, show that all test statistics are positive, with most being significant at the 1 % level. This further confirms that the Transformer model outperforms the other models in predicting realized volatility.

Table 3
Prediction accuracies and rankings of various models for realized volatility.

		MAE	MAPE	MSE	MSLE	SMAPE	Ranking
DL	**GRU_Pos_Neg**	0.3869(6)	0.7193(6)	0.3652(9)	0.0748(8)	0.5325(9)	8
	GRU_Sentiment	0.3913(8)	0.7454(8)	0.3685(10)	0.0762(10)	0.5383(10)	10
	LSTM_Pos_Neg	0.3806(5)	0.7070(5)	0.3524(5)	0.0732(5)	0.5232(7)	4
	LSTM_Sentiment	0.3882(7)	0.7310(7)	0.3629(7)	0.0753(9)	0.5324(8)	9
	Transformer_Pos_Neg	0.3609(1)	0.6404(1)	0.3248(1)	0.0654(1)	0.4979(1)	1
	Transformer_Sentiment	0.3661(2)	0.6994(4)	0.3321(2)	0.0664(2)	0.5072(3)	2
	WaveNet_Pos_Neg	0.3699(3)	0.6436(2)	0.3558(6)	0.0726(4)	0.5088(5)	3
	WaveNet_Sentiment	0.3767(4)	0.6494(3)	0.3650(8)	0.0745(7)	0.5163(6)	5
ML	**ABR_Pos_Neg**	2.4325(19)	5.9052(19)	6.4245(19)	0.9186(19)	1.3283(19)	19
	ABR_Sentiment	2.6182(20)	6.3153(20)	7.9284(20)	0.9985(20)	1.3482(20)	20
	DTR_Pos_Neg	0.6282(14)	1.1412(13)	1.0517(17)	0.1573(14)	0.6477(14)	14
	DTR_Sentiment	0.6314(15)	1.2162(14)	1.0128(15)	0.1687(15)	0.6431(13)	15
	GBR_Pos_Neg	0.6075(13)	1.4205(15)	0.6567(12)	0.1398(13)	0.6764(15)	13
	GBR_Sentiment	0.7293(16)	1.7257(17)	1.0265(16)	0.1851(16)	0.7399(17)	16
	KNR_Pos_Neg	0.4907(11)	0.9253(11)	0.6178(11)	0.1144(11)	0.5566(11)	11
	KNR_Sentiment	0.5299(12)	0.9922(12)	0.8756(13)	0.1216(12)	0.5663(12)	12
	SVR_Pos_Neg	0.3970(9)	0.7825(9)	0.3398(3)	0.0724(3)	0.5081(4)	6
	SVR_Sentiment	0.3970(10)	0.7846(10)	0.3435(4)	0.0733(6)	0.5063(2)	7
	XGBR_Pos_Neg	0.7514(17)	1.8154(18)	0.9507(14)	0.1996(17)	0.7437(18)	17
	XGBR_Sentiment	0.7847(18)	1.6596(16)	1.4471(18)	0.2210(18)	0.7141(16)	18

Note: *GRU*: Gate Recurrent Unit; *LSTM*: Long Short-Term Memory; *ABR*: AdaBoost Regressor; *DTR*: Decision Tree Regressor; *GBR*: Gradient Boosting Regressor; *KNR*: K-Nearest Neighbors Regressor; *SVR*: Support Vector Regression; *XGBR*: Extreme Gradient Boosting Regressor.

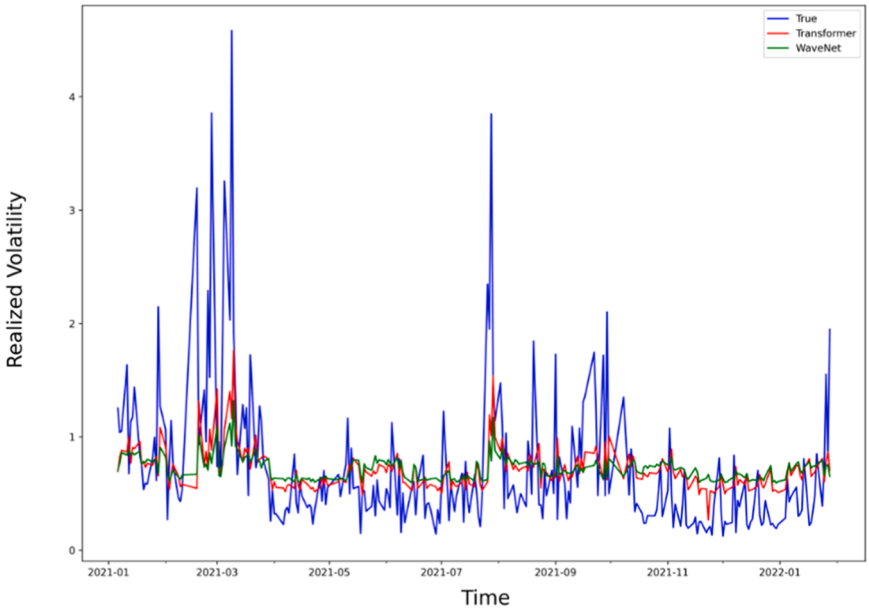

Fig. 3. The comparison of RV between transformer and WaveNet.

Table 4
The importance of sentiment-related features.

		Sentiment_Pos	Sentiment_Neg	Sentiment
Transformer	**MAE**	2.2900 %	3.6800 %	1.1700 %
	MAPE	4.6200 %	6.8800 %	2.8500 %
	MSE	2.5800 %	3.8700 %	0.6100 %
	MSLE	3.5300 %	5.8300 %	1.2400 %
	SMAPE	2.0700 %	3.2100 %	1.1700 %

Table 5
The DM test between transformer and others.

		Transformer		
		MAE	MAPE	MSE
DL	GRU	5.1000 ***	6.5374 ***	3.0667 ***
	LSTM	4.3339 ***	6.6843 ***	2.9895 ***
	WaveNet	1.8570 *	0.297	3.0182 ***
ML	ABR	32.8903 ***	22.0815 ***	18.6607 ***
	DTR	6.4214 ***	5.2715 ***	4.6087 ***
	GBR	6.7936 ***	10.8214 ***	3.7222 ***
	KNR	3.9399 ***	4.1342 ***	3.1125 ***
	SVR	1.8948 *	3.9152 ***	0.3857
	XGBR	8.7021 ***	10.3888 ***	5.3690 ***

Note: *, **, *** denote 10 %, 5 %, and 1 % significance levels, respectively.

Table 6
Five-step prediction accuracies and rankings of various models for realized volatility.

		MAE	MAPE	MSE	MSLE	SMAPE	Ranking
DL	GRU_Pos_Neg	0.3820(7)	0.7102(7)	0.3765(7)	0.0762(5)	0.5266(6)	7
	GRU_Sentiment	0.3888(8)	0.7335(8)	0.3762(6)	0.0770(7)	0.5377(8)	8
	LSTM_Pos_Neg	0.3781(5)	0.6884(5)	0.3795(8)	0.0766(6)	0.5181(5)	6
	LSTM_Sentiment	0.3814(6)	0.7086(6)	0.3648(3)	0.0745(3)	0.5271(7)	5
	Transformer_Pos_Neg	0.3708(1)	0.6822(4)	0.3583(1)	0.0724(2)	0.5107(1)	1
	Transformer_Sentiment	0.3713(2)	0.6812(3)	0.3583(2)	0.0721(1)	0.5123(4)	2
	WaveNet_Pos_Neg	0.3721(4)	0.6282(1)	0.3745(5)	0.0749(4)	0.5122(3)	3
	WaveNet_Sentiment	0.3720(3)	0.6367(2)	0.3735(4)	0.0770(8)	0.5116(2)	4
ML	ABR_Pos_Neg	3.4232(19)	7.6208(19)	14.6463(19)	1.3170(19)	1.4167(19)	19
	ABR_Sentiment	3.9132(20)	8.7724(20)	19.4122(20)	1.5289(20)	1.4392(20)	20
	DTR_Pos_Neg	1.6618(16)	3.8786(15)	5.8592(15)	0.5921(15)	0.9676(13)	15
	DTR_Sentiment	1.6553(15)	3.9910(16)	5.9157(16)	0.6048(16)	0.9732(14)	16
	GBR_Pos_Neg	1.4828(14)	3.7071(14)	3.0157(14)	0.5024(14)	1.0720(16)	14
	GBR_Sentiment	1.4041(13)	3.5768(13)	2.6381(13)	0.4712(13)	1.0574(15)	13
	KNR_Pos_Neg	0.6892(12)	1.3519(12)	1.3281(12)	0.1897(12)	0.6332(11)	12
	KNR_Sentiment	0.6173(11)	1.1827(11)	0.8966(11)	0.1532(11)	0.6349(12)	11
	SVR_Pos_Neg	0.4353(9)	0.8721(10)	0.4268(9)	0.0891(9)	0.5468(9)	9
	SVR_Sentiment	0.4353(10)	0.8694(9)	0.4273(10)	0.0894(10)	0.5478(10)	10
	XGBR_Pos_Neg	2.1832(18)	5.7402(18)	7.6979(18)	0.8562(18)	1.1321(18)	18
	XGBR_Sentiment	1.9554(17)	5.1062(17)	6.1322(17)	0.7437(17)	1.1033(17)	17

Note: *GRU*: Gate Recurrent Unit; *LSTM*: Long Short-Term Memory; *ABR*: AdaBoost Regressor; *DTR*: Decision Tree Regressor; *GBR*: Gradient Boosting Regressor; *KNR*: K-Nearest Neighbors Regressor; *SVR*: Support Vector Regression; *XGBR*: Extreme Gradient Boosting Regressor.

3.4. Multi-step forecasting

We conducted a five-step volatility forecast using a direct multi-step approach, which is simpler, more efficient, and prevents error accumulation compared to the one-step method. Results in Table 6 show that sentiment feature decomposition improves prediction accuracy for most models. The Transformer model continues to outperform others in overall performance.

3.5. VaR calculation

In previous sections, we used deep learning models with sentiment features to forecast realized volatility, aiding quantitative risk management. However, these forecasts do not directly capture potential losses. To address this, we introduce VaR, a predictive measure that accounts for extreme future events. As risk management evolves, VaR methods have advanced from basic statistical models to approaches like Monte Carlo simulations and the Peaks Over Threshold (POT) method based on extreme value theory in Appendix E.

Fig. 4 shows the 95 % confidence level VaR values for the optimal Transformer model with decomposed sentiment features, alongside daily log returns. During significant fluctuations in March 2021, VaR effectively served as a safety net, providing critical information for risk managers to mitigate potential losses.

Table 7 confirms that all p-values across different tests exceed the 5 % significance level, indicating that the VaR estimates pass all backtesting criteria and are reliable for risk forecasting.

4. Conclusion

This paper combines sentiment analysis with deep learning to forecast realized volatility in financial markets. Using a BERT-based

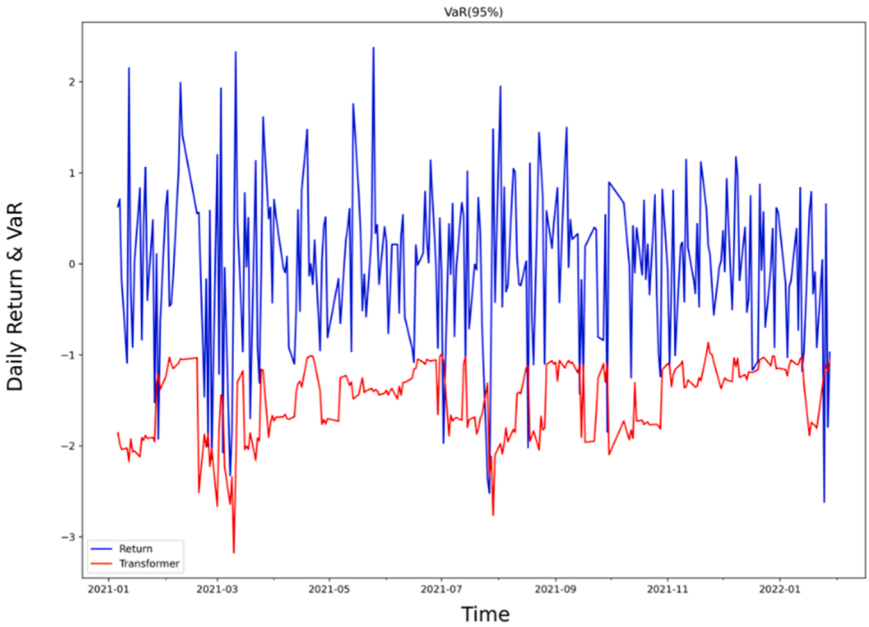

Fig. 4. VaR and return based on transformer model.

Table 7
The statistical test for VaR.

	VaR_lvl	num_hits	pct_hits	PVuc	UC	PVcci
Transformer	0.05	15	0.0577	0.5781	Accept	0.8836
	CCI	**PVcc**	**CC**	**PVdt**	**DT**	**tick_loss**
	Accept	0.8476	Accept	0.2400	Accept	0.1051

Note: *VaR_lvl*: the desired VaR significance level; *num_hits*: number of hits; *pct_hits*: percentage of hits; *PVuc*: p-value of the unconditional coverage test; *UC*: unconditional coverage test (Christoffersen, 1998); *PVcci*: p-value of the conditional coverage independence test; *CCI*: conditional coverage independence test (Christoffersen, 1998); *PVcc*: p-value of the conditional coverage test; *CC*: conditional coverage test (Christoffersen, 1998); P*Vdt*: p-value of the duration test; *DT*: duration test (Christoffersen and Pelletier, 2004); *tick_loss*: tick loss function.

model to extract positive and negative sentiment indices from stock forums, and combining these indices with traditional technical indicators, we build a comprehensive set of features for improved volatility predictions. The Transformer model effectively utilizes these features, resulting in improved prediction accuracy. The results show that negative sentiment has a stronger impact on realized volatility, and separating positive and negative sentiment features improves accuracy compared to using a single sentiment index. This approach also enhances VaR forecasting.

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CRediT authorship contribution statement

Yuping Song: Writing – original draft, Project administration, Funding acquisition. **Yilun Zhang:** Software, Investigation, Formal analysis. **Jiefei Huang:** Validation, Resources, Data curation. **Aijun Yang:** Writing – review & editing, Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.frl.2025.108210.

Data availability

Data will be made available on request.

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